Methanol

24229-2.5L

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1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product information

Trade name	:	Methanol
Number	:	000000020240
Recommended use of the chemical and restrictions on use	:	Laboratory chemicals
Manufacturer or supplier's details	:	Honeywell Specialty Chemicals Seelze GmbH Wunstorfer Straße 40 Seelze, 30926
For further information, please contact:	:	1-800-368-0050 +1-231-726-3171 (Monday-Friday, 9:00am-5:00pm)
In case of emergency call	:	**Medical: 1-800-498-5701 or +1-303-389-1414**
	:	**Transportation (CHEMTREC): 1-800-424-9300 or +1-703-527-3887** **In Vietnam: +(84)-838012436**
	:	(24 hours/day, 7 days/week)

2. HAZARDS IDENTIFICATION

Classification of the substance or mixture

Classification of the substance or mixture	:	Flammable liquids, Category 2 Acute toxicity, Category 5, Oral Acute toxicity, Category 5, Inhalation Acute toxicity, Category 5, Dermal Specific target organ toxicity - single exposure, Category 1, Eyes, Nervous system, Systemic toxicity

GHS Label elements, including precautionary statements

Symbol(s) :



Signal word	:	Danger
Hazard statements	:	Highly flammable liquid and vapour.

May be harmful if swallowed.
May be harmful in contact with skin.
May be harmful if inhaled.
Causes damage to organs.

Precautionary statements : **Prevention:**
Keep away from heat/ sparks/ open flames/ hot surfaces. No smoking.
Keep container tightly closed.
Ground/bond container and receiving equipment.
Use explosion-proof electrical/ ventilating/ lighting equipment.
Use only non-sparking tools.
Take precautionary measures against static discharge.
Do not breathe dust/ fume/ gas/ mist/ vapours/ spray.
Wash skin thoroughly after handling.
Do not eat, drink or smoke when using this product.
Wear protective gloves/protective clothing/eye protection/face protection.

Response:
IF ON SKIN (or hair): Remove/ Take off immediately all contaminated clothing. Rinse skin with water/ shower.
IF exposed: Call a POISON CENTER or doctor/ physician.
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam for extinction.

Storage:
Store in a well-ventilated place. Keep cool.
Store locked up.

Disposal:
Dispose of contents/ container to an approved waste disposal plant.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Formula : CH_4O

Chemical nature : Substance

Chemical name	CAS-No.	Concentration
Methanol Methanol	67-56-1	<=100.00 %

4. FIRST AID MEASURES

General advice	:	First aider needs to protect himself. Move out of dangerous area. Take off all contaminated clothing immediately. Show this safety data sheet to the doctor in attendance.
Inhalation	:	Call a physician immediately. Remove to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Use oxygen as required, provided a qualified operator is present.
Skin contact	:	After contact with skin, wash immediately with plenty of water. Call a physician immediately.
Eye contact	:	Rinse thoroughly with plenty of water, also under the eyelids. Protect unharmed eye. Call a physician immediately.
Ingestion	:	Immediately give large quantities of water to drink. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Call a physician immediately.
Notes to physician	:	Treat symptomatically.

5. FIREFIGHTING MEASURES

Suitable extinguishing media	:	Water spray Alcohol-resistant foam Carbon dioxide (CO2) Dry powder
Unsuitable extinguishing media	:	Do not use a solid water stream as it may scatter and spread fire.
Specific hazards during firefighting	:	Flammable. Vapours may form explosive mixtures with air. Vapours are heavier than air and may spread along floors. Vapors may travel to areas away from work site before igniting/flashing back to vapor source. In case of fire hazardous decomposition products may be produced such as: Carbon monoxide Carbon dioxide (CO2) Formaldehyde
Special protective equipment	:	Wear self-contained breathing apparatus and protective suit.

for firefighters

Further information : Use water spray to cool unopened containers.
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures : Ensure adequate ventilation.
Remove all sources of ignition.
Wear personal protective equipment. Unprotected persons must be kept away.

Environmental precautions : Prevent further leakage or spillage if safe to do so.
Prevent product from entering drains.
Discharge into the environment must be avoided.
Do not flush into surface water or sanitary sewer system.
Do not allow run-off from fire fighting to enter drains or water courses.

Methods and materials for containment and cleaning up : Ventilate the area.
No sparking tools should be used.
Use explosion-proof equipment.
Contain spillage, soak up with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and transfer to a container for disposal according to local / national regulations (see section 13).

7. HANDLING AND STORAGE

Handling

Precautions for safe handling : Exhaust ventilation at the object is necessary.
Use explosion-proof equipment.
Wear suitable protective clothing and gloves.

Advice on protection against fire and explosion : The heavy vapours can overcome a considerable distance up to the source of ignition.
Use only in explosion-proof areas.
Keep product and empty container away from heat and sources of ignition.
Take necessary action to avoid static electricity discharge (which might cause ignition of organic vapours).

Temperature class : T1

Storage

Conditions for safe storage, including any incompatibilities	:	Keep only in the original container, tightly closed, in a well ventilated place. Store at room temperature. (Ambient temperature: > 0 < 35°C) Protect from atmospheric moisture and water.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Components	CAS-No.	Value	Control parameters	Update	Basis
Methanol Methanol	67-56-1	TWA : Time weighted average (TWA):	50 mg/m3	06 2019	VN OEL:Vietnam. OELs, as amended
		STEL : Short-term exposure limit (STEL):	100 mg/m3	06 2019	VN OEL:Vietnam. OELs, as amended

Appropriate engineering controls

Use with local exhaust ventilation.
Electrical equipment should be protected to the appropriate standard.

Individual protection measures, such as personal protective equipment

Respiratory protection	:	In the case of vapour formation use a respirator with an approved filter. Recommended Filter type: Organic gas and low boiling vapour type
Hand protection	:	Impervious butyl rubber gloves Gloves must be inspected prior to use. Replace when worn.
Eye protection	:	Safety goggles
Skin and body protection	:	Wear as appropriate: Solvent-resistant apron Flame retardant antistatic protective clothing. If splashes are likely to occur, wear: Protective suit

Hygiene measures	:	When using do not eat, drink or smoke. Wash hands before breaks and immediately after handling the product. Keep working clothes separately. Do not swallow. Do not breathe vapours or spray mist. Avoid contact with skin, eyes and clothing.
Protective measures	:	Ensure that eyewash stations and safety showers are close to the workstation location. Legal requirements are to be considered in regard of the selection, use and care of personal protective equipment. Do not swallow. Do not breathe vapours or spray mist.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	:	liquid
Colour	:	colourless
Odour	:	characteristic
pH	:	Note: Not applicable
Melting point/range	:	-98 °C
Boiling point/boiling range	:	64 - 65 °C at 1,013 hPa
Flash point	:	52 °F (11 °C) Method: DIN 51755
Evaporation rate	:	Note: No data available
Flammability	:	Not applicable (solid, gas)
Lower explosion limit	:	5.5 %(V)
Upper explosion limit	:	50 %(V)

Vapour pressure	:	128 hPa at 20 °C(68 °F) 532 hPa at 50 °C(122 °F)
Vapour density	:	Note: No data available
Density	:	0.79 g/cm3 at 20 °C
Water solubility	:	Note: completely miscible
Partition coefficient: n-octanol/water	:	log Pow: -0.71
Ignition temperature	:	455 °C Method: DIN 51794
Auto-ignition temperature	:	Note: not auto-flammable
Decomposition temperature	:	Note: At normal pressure may be distilled without decomposition.
Viscosity, dynamic	:	ca. 0.55 mPa.s at 25 °C
Viscosity, kinematic	:	Note: No data available
Oxidizing properties	:	The substance or mixture is not classified as oxidizing.
Molecular weight	:	32.04 g/mol

10. STABILITY AND REACTIVITY

Chemical stability	:	Stable under recommended storage conditions.
Possibility of hazardous reactions	:	Hazardous polymerisation does not occur.
Conditions to avoid	:	Heat, flames and sparks. Keep away from direct sunlight.

Incompatible materials to avoid	:	Oxidizing agents
Hazardous decomposition products	:	In case of fire hazardous decomposition products may be produced such as: Carbon monoxide Carbon dioxide (CO2) Formaldehyde

11. TOXICOLOGICAL INFORMATION

Acute oral toxicity	:	LD50: 5,630 mg/kg Species: Rat
Acute inhalation toxicity	:	LC50: > 80 mg/l Exposure time: 4 h Species: Rat
Acute dermal toxicity	:	LD50: > 10,000 mg/kg Species: Rabbit
Skin irritation	:	Species: Rabbit Result: No skin irritation
Eye irritation	:	Species: Rabbit Result: No eye irritation
Sensitisation	:	Note: No data available
Repeated dose toxicity	:	Species: Rat Application Route: Inhalation Note: Developmental Toxicity NOAEL (maternal toxicity) 10,000 ppm NOAEL (developmental toxicity) 5,000 ppm Skeletal and visceral malformations.
Genotoxicity in vitro	:	Note: In vitro tests did not show mutagenic effects
Genotoxicity in vivo	:	Note: In vivo tests did not show mutagenic effects

Further information	:	Note: When swallowed, there is a danger of blindness. Solvent vapours have a narcotic effect if inhaled in high concentrations. Causes damage to organs (eyes, nervous system, systemic toxicity)

12. ECOLOGICAL INFORMATION

Toxicity

Toxicity to fish	:	LC50: 15,400 mg/l Exposure time: 96 h Species: Lepomis macrochirus (Bluegill sunfish)
Toxicity to daphnia and other aquatic invertebrates	:	EC50: > 10,000 mg/l Exposure time: 48 h Species: Daphnia magna (Water flea)
Toxicity to algae	:	Note: No data available
Toxicity to bacteria	:	EC50: ca. 71,000 mg/l Species: Bacteria

Persistence and degradability

Biodegradability	:	Result: Readily biodegradable. Value: 99 % Method: OECD Test Guideline 301D

Bioaccumulative potential

Bioaccumulation	:	Note: Bioaccumulation is unlikely.

Mobility in soil

Mobility	:	Note: No data available

Other adverse effects

Additional ecological information	:	Do not flush into surface water or sanitary sewer system.

Biochemical Oxygen Demand (BOD)	:	Value: 1,120 mg/g
Chemical Oxygen Demand (COD)	:	Value: 1,500 mg/g

13. DISPOSAL CONSIDERATIONS

Disposal methods	:	In accordance with local and national regulations.

14. TRANSPORT INFORMATION

ADR

UN/ID No.	:	UN 1230
Description of the goods	:	METHANOL
Class	:	3
Packing group	:	II
Classification Code	:	FT1
Hazard Identification Number	:	336
Labels	:	3 (6.1)

IATA

UN/ID No.	:	UN 1230
Description of the goods	:	Methanol
Class	:	3
Packing group	:	II
Labels	:	3 (6.1)
Packing instruction (cargo aircraft)	:	364
Packing instruction (passenger aircraft)	:	352
Packing instruction (passenger aircraft)	:	Y341

IMDG

UN/ID No.	:	UN 1230
Description of the goods	:	METHANOL
Class	:	3
Packing group	:	II
Labels	:	3 (6.1)
EmS Number 1	:	F-E
EmS Number 2	:	S-D
Marine pollutant	:	no

15. REGULATORY INFORMATION

National regulatory information

: No information available.

Other international regulations

Notification status

US. Toxic Substances Control Act	:	On TSCA Inventory
Australia. Industrial Chemical (Notification and Assessment) Act	:	On the inventory, or in compliance with the inventory
Canada. Canadian Environmental Protection Act (CEPA). Domestic Substances List (DSL)	:	All components of this product are on the Canadian DSL
Japan. Kashin-Hou Law List	:	On the inventory, or in compliance with the inventory
Korea. Existing Chemicals Inventory (KECI)	:	On the inventory, or in compliance with the inventory
Philippines. Inventory of Chemicals and Chemical Substances (PICCS)	:	On the inventory, or in compliance with the inventory
China. Inventory of Existing Chemical Substances (IECSC)	:	On the inventory, or in compliance with the inventory
New Zealand. Inventory of Chemicals (NZIoC), as published by ERMA New Zealand	:	On the inventory, or in compliance with the inventory

16. OTHER INFORMATION

		HMIS III	NFPA
Health hazard	:	2*	1
Flammability	:	3	3
Physical Hazard	:	0	
Instability	:		0

* - Chronic health hazard
Hazard rating and rating systems (e.g. HMIS® III, NFPA): This information is intended solely for the use of individuals trained in the particular system.

Further information

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text. Final determination of suitability of any material is the sole responsibility of the user. This information should not constitute a guarantee for any specific product properties.

Changes since the last version are highlighted in the margin. This version replaces all previous versions.
Prepared by Honeywell Performance Materials and Technologies Product Stewardship Group